

Newman-Penrose Formalism Calculations

Stephani et al., Equation (15.23)

Metric

$$ds^2 = \left(\frac{2m-r}{r} \right) dt \otimes dt + \left(-\frac{2m+r}{r} \right) dt \otimes dr + \left(-\frac{2m+r}{r} \right) dr \otimes dt + \left(\frac{4(m^2+mr)}{2mr-r^2} \right) dr \otimes dr + r^2 d\vartheta \otimes d\vartheta + r^2 \sin(\vartheta)^2 d\varphi \otimes d\varphi$$

Coordinates

$$(x^\mu) = (t, r, \vartheta, \varphi)$$

Metric Functions

$$u = 2m \log(-2m+r) + t$$

Null Tetrad

Covariant components

$$l = (1) dt + \left(-\frac{2m}{2m-r} \right) dr$$

$$n = \left(-\frac{m}{r} + \frac{1}{2} \right) dt + \left(-\frac{m}{2m-r} + \frac{2m^2}{(2m-r)r} + 1 \right) dr$$

$$m = \left(\frac{1}{2} \sqrt{2} r \right) d\vartheta + \left(\frac{1}{2} i \sqrt{2} r \sin(\vartheta) \right) d\varphi$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2} r \right) d\vartheta + \left(-\frac{1}{2} i \sqrt{2} r \sin(\vartheta) \right) d\varphi$$

Contravariant components

$$l = \left(-\frac{2m}{2m-r} \right) dt + (-1) dr$$

$$n = \left(-\frac{m}{r} - 1 \right) dt + \left(-\frac{m}{r} + \frac{1}{2} \right) dr$$

$$m = \left(\frac{\sqrt{2}}{2r} \right) d\vartheta + \left(\frac{i\sqrt{2}}{2r\sin(\vartheta)} \right) d\varphi$$

$$\bar{m} = \left(\frac{\sqrt{2}}{2r} \right) d\vartheta + \left(-\frac{i\sqrt{2}}{2r\sin(\vartheta)} \right) d\varphi$$

Directional Derivatives

$$D = -\frac{2m}{2m-r} \partial_t - \partial_r$$

$$\Delta = -\frac{m}{r} - 1 \partial_t - \frac{m}{r} + \frac{1}{2} \partial_r$$

$$\delta = \frac{\sqrt{2}}{2r} \partial_\vartheta + \frac{i\sqrt{2}}{2r\sin(\vartheta)} \partial_\varphi$$

$$\bar{\delta} = \frac{\sqrt{2}}{2r} \partial_\vartheta - \frac{i\sqrt{2}}{2r\sin(\vartheta)} \partial_\varphi$$

Spin Coefficients

$$\rho = \frac{1}{r}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = -\frac{m}{r^2} + \frac{1}{2r}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = -\frac{\sqrt{2} \cos(\vartheta)}{4r \sin(\vartheta)}$$

$$\beta = \frac{\sqrt{2} \cos(\vartheta)}{4r \sin(\vartheta)}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\gamma = -\frac{m}{2r^2}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = 0$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = 0$$

$$\Lambda = 0$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = -\frac{m}{r^3}$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"